Worksheet 1.2 — Solving Linear Systems by Graphing

Grade 10 Academic (MPM2D) · Unit 1: Linear Systems

The Big Ideas

A **linear system** is two lines considered together; its **solution** is the point (x, y) that lies on *both* lines — the point of intersection (POI). Solve by graphing: draw the first line, draw the second line on the same grid, then read where they cross.

- For $y = mx + b$: plot the y-intercept b, use the slope's $\frac{\text{rise}}{\text{run}}$ to find a second point, **check** that point, then draw the line.
- For standard form $Ax + By = C$: find the x-intercept (set $y = 0$) and y-intercept (set $x = 0$) and draw the line through them — no need to solve for y first.
- The POI must satisfy *both* equations — always **check** by substituting back.
- **One solution:** the lines cross once (different slopes).
- **No solution:** parallel lines (equal slopes, different y-intercepts).
- **Infinitely many:** the same line (equal slopes *and* intercepts).

Concept Spotlight

What a system is

A **linear system** is two lines considered together. Its **solution** is the point (x, y) that lies on *both* lines the coordinates that satisfy both equations at once.

Reading the intersection

Graph both lines on the same grid; where they cross is the solution. Always check by substituting that point back into both equations.

Its limits

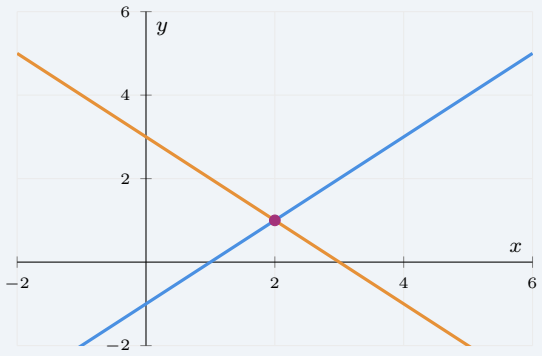
Graphing shows the idea clearly, but it is only as accurate as your reading of the grid fractional or large solutions call for an algebraic method.

Worked Examples

Example 1 — Checking a solution

Verify that $(2, 1)$ solves the system $y = x - 1$ and $y = -x + 3$.

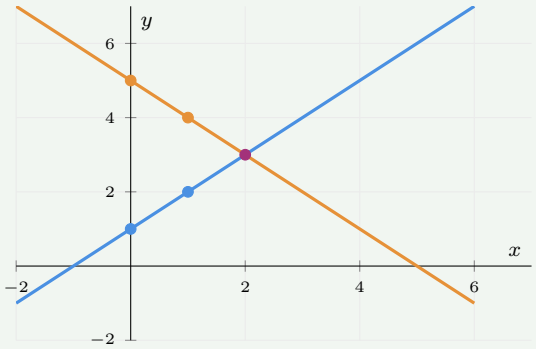
Solution. Substitute the point into *both* equations. First equation: $y = x - 1 \Rightarrow 1 = 2 - 1 = 1$ ✓. Second equation: $y = -x + 3 \Rightarrow 1 = -2 + 3 = 1$ ✓. Since it makes both equations true, $(2, 1)$ is the point where the two lines cross.



Example 2 — Graphing a simple system

Solve by graphing: $y = x + 1$ and $y = -x + 5$.

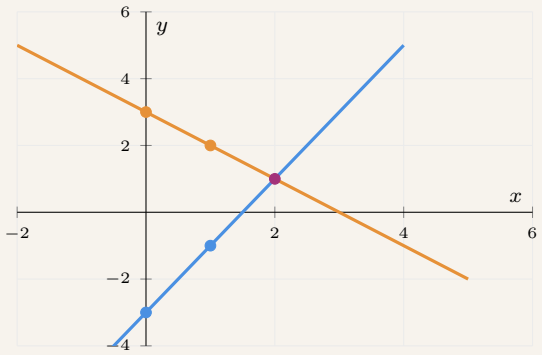
Solution. **Step 1 (draw the first line, $y = x + 1$):** $b = 1$, plot $(0, 1)$. Slope $\frac{1}{1}$ (rise 1, run 1): from $(0, 1)$ go right 1, up 1, to $(1, 2)$. Check: $y = 1 + 1 = 2$ ✓. Draw the line through both points. **Step 2 (draw the second line, $y = -x + 5$):** $b = 5$, plot $(0, 5)$. Slope $\frac{-1}{1}$: from $(0, 5)$ go right 1, down 1, to $(1, 4)$. Check: $y = -1 + 5 = 4$ ✓. Draw the line. **Step 3 (find the intersection):** the two lines cross at $(2, 3)$. **Step 4 (check):** $2 + 1 = 3$ ✓ and $-2 + 5 = 3$ ✓. Solution $(2, 3)$.



Example 3 — Steeper lines

Solve by graphing: $y = 2x - 3$ and $y = -x + 3$.

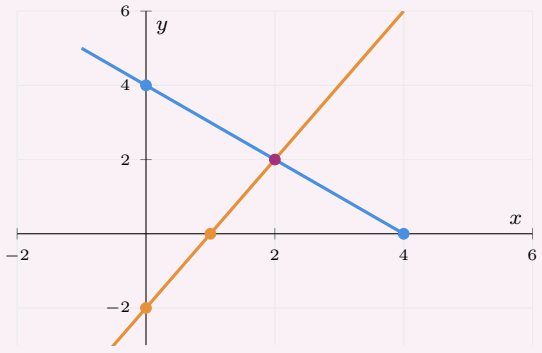
Solution. **Step 1 (draw the first line, $y = 2x - 3$):** $b = -3$, plot $(0, -3)$. Slope $\frac{2}{1}$: from $(0, -3)$ go right 1, up 2, to $(1, -1)$. Check: $y = 2(1) - 3 = -1$ ✓. Draw the line. **Step 2 (draw the second line, $y = -x + 3$):** $b = 3$, plot $(0, 3)$. Slope $\frac{-1}{1}$: from $(0, 3)$ go right 1, down 1, to $(1, 2)$. Check: $y = -1 + 3 = 2$ ✓. Draw the line. **Step 3 (find the intersection):** extending both lines, they cross at $(2, 1)$. Confirm algebraically: $2x - 3 = -x + 3 \Rightarrow 3x = 6 \Rightarrow x = 2$, so $y = 1$. **Step 4 (check):** $1 = 2(2) - 3$ ✓ and $1 = -2 + 3$ ✓. Solution $(2, 1)$.



Example 4 — One equation in standard form

Solve by graphing: $x + y = 4$ and $y = 2x - 2$.

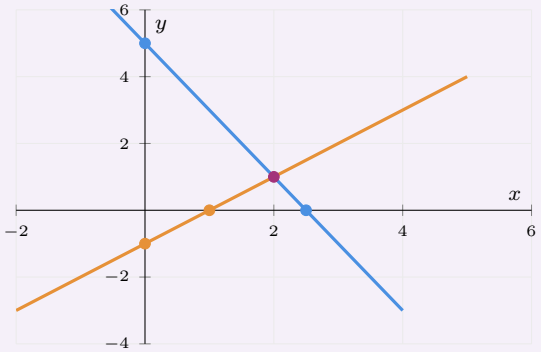
Solution. **Step 1 (draw the first line, $x + y = 4$, standard form):** x-intercept (set $y = 0$): $x = 4$, plot $(4, 0)$. y-intercept (set $x = 0$): $y = 4$, plot $(0, 4)$. Draw the line through both intercepts — no need to solve for y first. **Step 2 (draw the second line, $y = 2x - 2$, slope-intercept form):** $b = -2$, plot $(0, -2)$. Slope $\frac{2}{1}$: from $(0, -2)$ go right 1, up 2, to $(1, 0)$. Check: $y = 2(1) - 2 = 0$ ✓. Draw the line. **Step 3 (find the intersection):** the lines cross at $(2, 2)$. Confirm: $x + y = 4 \Rightarrow 2 + 2 = 4$ ✓; also $-x + 4 = 2x - 2 \Rightarrow 6 = 3x \Rightarrow x = 2$. **Step 4 (check):** $2 + 2 = 4$ ✓ and $2 = 2(2) - 2$ ✓. Solution $(2, 2)$.



Example 5 — Both in standard form

Solve by graphing: $2x + y = 5$ and $x - y = 1$.

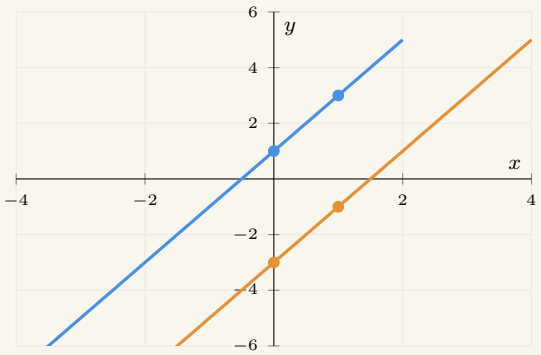
Solution. **Step 1 (draw the first line, $2x + y = 5$):** x-intercept (set $y = 0$): $2x = 5 \Rightarrow x = 2.5$, plot $(2.5, 0)$. y-intercept (set $x = 0$): $y = 5$, plot $(0, 5)$. Draw the line through both. **Step 2 (draw the second line, $x - y = 1$):** x-intercept (set $y = 0$): $x = 1$, plot $(1, 0)$. y-intercept (set $x = 0$): $-y = 1 \Rightarrow y = -1$, plot $(0, -1)$. Draw the line through both. **Step 3 (find the intersection):** the lines cross at $(2, 1)$. Confirm by adding the equations: $2x + y = 5$ and $x - y = 1$ give $3x = 6 \Rightarrow x = 2$, so $y = 1$. **Step 4 (check):** $2(2) + 1 = 5$ ✓ and $2 - 1 = 1$ ✓. Solution $(2, 1)$.



Example 6 — Parallel lines (no solution)

Solve by graphing: $y = 2x + 1$ and $y = 2x - 3$.

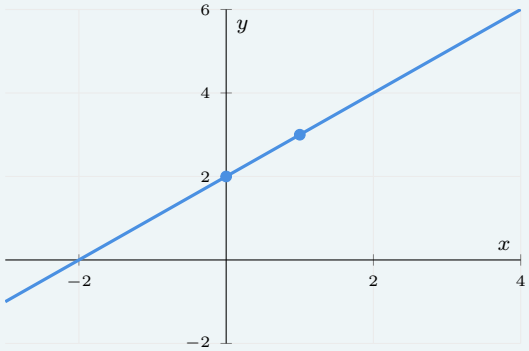
Solution. **Step 1 (draw the first line, $y = 2x + 1$):** $b = 1$, plot $(0, 1)$. Slope $\frac{2}{1}$: from $(0, 1)$ go right 1, up 2, to $(1, 3)$. Check: $y = 2(1) + 1 = 3$ ✓. Draw the line. **Step 2 (draw the second line, $y = 2x - 3$):** $b = -3$, plot $(0, -3)$. Slope $\frac{2}{1}$ — *the same rise and run as the first line*: from $(0, -3)$ go right 1, up 2, to $(1, -1)$. Check: $y = 2(1) - 3 = -1$ ✓. Draw the line. **Step 3 (find the intersection):** both lines rise at the identical rate, so the second line stays exactly 4 units below the first everywhere — they never meet, no matter how far extended. **Conclusion:** equal slopes, different y-intercepts ⇒ parallel lines ⇒ **no solution**.



Example 7 — Same line (infinitely many)

Solve: $y = x + 2$ and $2x - 2y = -4$.

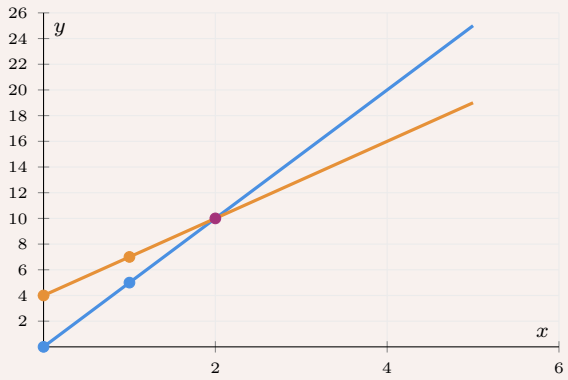
Solution. **Step 1 (simplify before drawing):** divide the second equation by 2: $2x - 2y = -4 \Rightarrow x - y = -2 \Rightarrow y = x + 2$ — identical to the first equation. **Step 2 (draw the line, $y = x + 2$):** $b = 2$, plot $(0, 2)$. Slope $\frac{1}{1}$: from $(0, 2)$ go right 1, up 1, to $(1, 3)$. Check: $y = 1 + 2 = 3$ ✓. Draw the line. **Step 3 (draw the "second" line):** since the simplified second equation is exactly $y = x + 2$ again, its y-intercept and second point land on the same two points — it traces over the first line exactly, with no separate crossing point. **Conclusion:** the two equations describe the *same* line, so there are **infinitely many solutions**.



Example 8 — An application

Gym A charges \$5 per visit: $C = 5v$. Gym B charges \$4 plus \$3 per visit: $C = 3v + 4$. When is the cost equal?

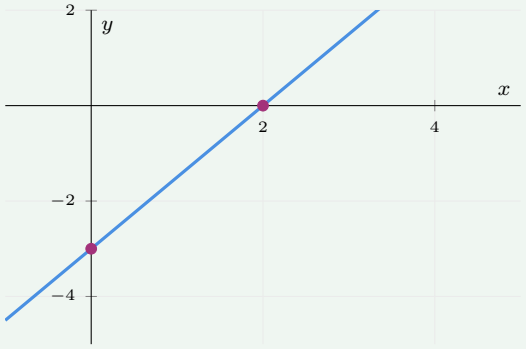
Solution. **Step 1 (draw the first line, $C = 5v$):** $b = 0$, plot $(0, 0)$. Slope $\frac{5}{1}$: from $(0, 0)$ go right 1 visit, up \$5, to $(1, 5)$. Check: $C = 5(1) = 5$ ✓. Draw the line. **Step 2 (draw the second line, $C = 3v + 4$):** $b = 4$, plot $(0, 4)$. Slope $\frac{3}{1}$: from $(0, 4)$ go right 1, up \$3, to $(1, 7)$. Check: $C = 3(1) + 4 = 7$ ✓. Draw the line. **Step 3 (find the intersection):** extending both lines, they cross at $(2, 10)$. Confirm: $5v = 3v + 4 \Rightarrow 2v = 4 \Rightarrow v = 2$ visits. **Step 4 (check):** $5(2) = 10$ ✓ and $3(2) + 4 = 10$ ✓. The costs are equal at $v = 2$ visits, $C = \$10$.



Example 9 — Solve and classify (harder)

Solve and classify: $3x - 2y = 6$ and $y = \frac{3}{2}x - 3$.

Solution. **Step 1 (draw the first line, $3x - 2y = 6$, standard form):** x-intercept (set $y = 0$): $3x = 6 \Rightarrow x = 2$, plot $(2, 0)$. y-intercept (set $x = 0$): $-2y = 6 \Rightarrow y = -3$, plot $(0, -3)$. Draw the line through both. **Step 2 (draw the second line, $y = \frac{3}{2}x - 3$, slope-intercept form):** $b = -3$, plot $(0, -3)$ — *the same point as line 1's y-intercept.* Slope $\frac{3}{2}$ (rise 3, run 2): from $(0, -3)$ go right 2, up 3, to $(2, 0)$ — *the same point as line 1's x-intercept.* **Step 3 (compare):** both lines pass through the exact same two points, $(0, -3)$ and $(2, 0)$, so they are the identical line. **Conclusion:** the system has **infinitely many solutions** (coincident lines).



Your Turn — Show Your Work

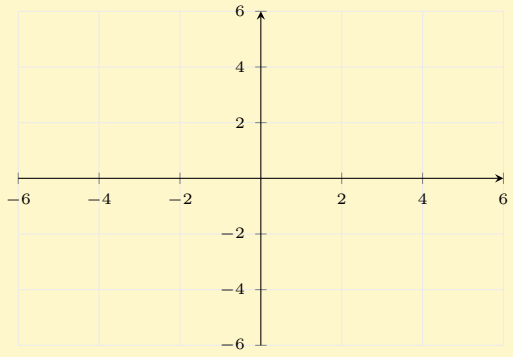
For each question, show all steps clearly in the space provided.

Question 1

Verify whether $(1, 4)$ is the solution of $y = 2x + 2$ and $y = -x + 5$.

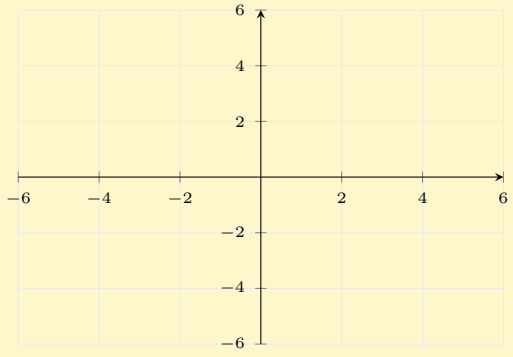
Question 2

Solve by graphing: $y = x - 2$ and $y = -x + 4$.



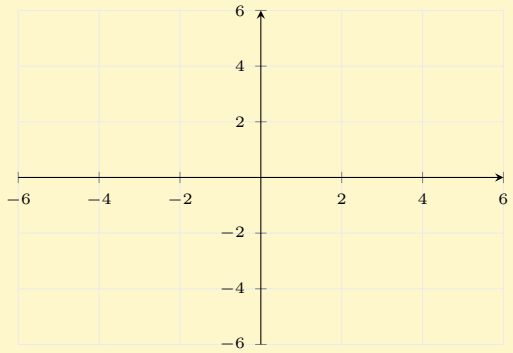
Question 3

Solve by graphing: $y = 2x + 1$ and $y = -x - 2$.



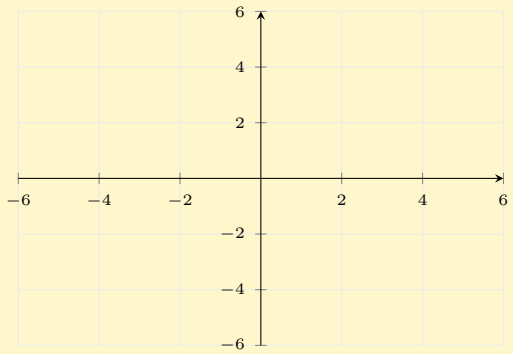
Question 4

Solve by graphing: $y = \frac{1}{2}x + 1$ and $y = -x + 4$.



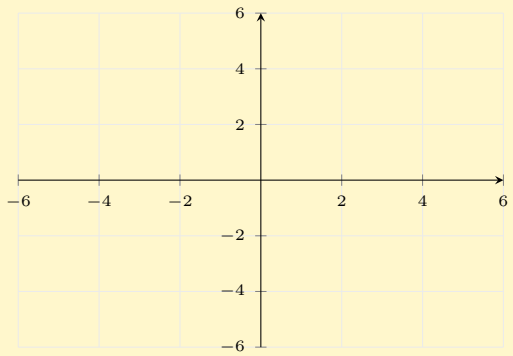
Question 5

Rewrite $x + y = 6$ and $y = 2x$ in $y = mx + b$ form, then solve by graphing.



Question 6

Solve by graphing: $2x + y = 3$ and $x - y = 3$.



Question 7

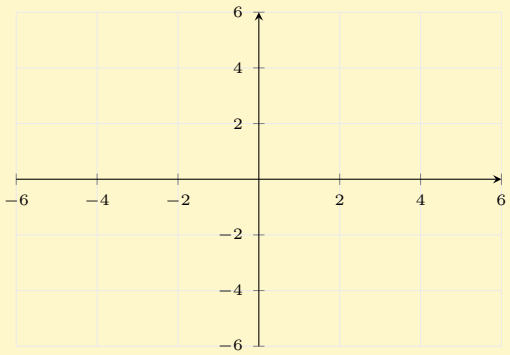
How many solutions does the system $y = 3x - 1$, $y = 3x + 4$ have? Explain.

Question 8

How many solutions does $y = -x + 2$, $3x + 3y = 6$ have? Explain.

Question 9

Solve by graphing: $y = -2x + 5$ and $y = x - 1$.



Question 10

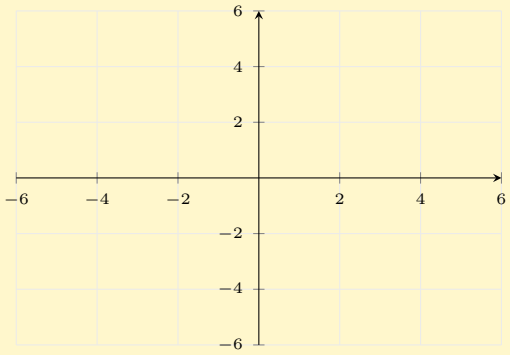
A taxi charges \$3 plus \$2/km: $C = 2d + 3$. Another charges \$5/km: $C = 5d$. When is the cost equal?

Question 11

Without graphing, decide whether $(4, 1)$ is the solution of $x + y = 5$ and $2x - y = 7$.

Question 12

Solve by graphing: $y = 4$ and $y = 2x - 2$.



Question 13 — Challenge

Two lines both have slope 2. One passes through $(0, 1)$; describe the number of solutions if the other passes through (a) $(0, 1)$ and (b) $(0, 5)$.

Question Recap & Answer Key

Question Recap

1. Verify whether $(1, 4)$ is the solution of $y = 2x+2$ and $y = -x + 5$.
2. Solve by graphing: $y = x - 2$ and $y = -x + 4$.
3. Solve by graphing: $y = 2x + 1$ and $y = -x - 2$.
4. Solve by graphing: $y = \frac{1}{2}x + 1$ and $y = -x + 4$.
5. Rewrite $x + y = 6$ and $y = 2x$ in $y = mx + b$ form, then solve by graphing.
6. Solve by graphing: $2x + y = 3$ and $x - y = 3$.
7. How many solutions does the system $y = 3x - 1$, $y = 3x + 4$ have? Explain.
8. How many solutions does $y = -x + 2$, $3x + 3y = 6$ have? Explain.
9. Solve by graphing: $y = -2x + 5$ and $y = x - 1$.
10. A taxi charges \$3 plus \$2/km: $C = 2d + 3$. Another charges \$5/km: $C = 5d$. When is the cost equal?
11. Without graphing, decide whether $(4, 1)$ is the solution of $x + y = 5$ and $2x - y = 7$.
12. Solve by graphing: $y = 4$ and $y = 2x - 2$.
13. Two lines both have slope 2. One passes through $(0, 1)$; describe the number of solutions if the other passes through (a) $(0, 1)$ and (b) $(0, 5)$.

✓ Answer Key

1. $2(1) + 2 = 4$ ✓, $-1 + 5 = 4$ ✓ — yes, $(1, 4)$ is the solution.
2. $(3, 1)$.
3. $(-1, -1)$.
4. $\frac{1}{2}x + 1 = -x + 4 \Rightarrow \frac{3}{2}x = 3 \Rightarrow x = 2$, $y = 2$: $(2, 2)$.
5. $y = -x + 6$, $y = 2x$; $2x = -x + 6 \Rightarrow x = 2$, $y = 4$: $(2, 4)$.
6. $y = -2x + 3$, $y = x - 3$; $x = 2$, $y = -1$: $(2, -1)$.
7. None — equal slopes, different intercepts (parallel).
8. Infinitely many — $3x+3y = 6$ simplifies to $y = -x+2$ (same line).
9. $-2x + 5 = x - 1 \Rightarrow x = 2$, $y = 1$: $(2, 1)$.
10. $2d + 3 = 5d \Rightarrow d = 1$ km, cost \$5.
11. $4 + 1 = 5$ ✓, $2(4) - 1 = 7$ ✓ — yes.
12. $2x - 2 = 4 \Rightarrow x = 3$: $(3, 4)$.
13. (a) same line $\Rightarrow$ infinitely many; (b) parallel $\Rightarrow$ no solution.